

Systems of linear equations

Two equations, two unknowns, and a solution that is a pair — the point where two lines cross.

LESSON 145–146

2 × 40 MIN

MATEMATIKA 7, §62

S8 2.5 EXTENSION

LESSON CLOCK

15'

20'

25'

20'

5'

One equation, two unknowns	15 min
What a system is	20 min
Solving by drawing	25 min
One, none, or infinitely many	20 min
Homework	5 min

LEARNING OBJECTIVES

- Recognise a linear equation in two unknowns and describe its solutions.
- Understand a system, and that its solution is an ordered pair.
- Solve a system graphically.
- Say when a system has one solution, none, or infinitely many.

TEXTBOOK

National	pp. 432–439
Cambridge	Stage 8, pp. 31–36
Workbook	Extension sheet 2D

01

Explanation

One equation, two unknowns

$x + y = 5$ has no single answer: $(1; 4)$, $(2; 3)$, $(0; 5)$ and infinitely many other pairs all satisfy it.

x	0	1	2	3	5
$y = 5 - x$	5	4	3	2	0

Plotted, those pairs lie on a straight line. A linear equation in two unknowns **is** a line.

WHY ONE EQUATION IS NOT ENOUGH

One condition on two unknowns leaves one degree of freedom, and the answers fill a line. To pin the pair down, a second independent condition is needed — which is the whole idea of a system.

What a system is

{ x + y = 5 ; x - y = 1 }

A **system** asks for the pairs that satisfy **both** equations at once. Its solution is written as an ordered pair.

PAIR	$X + Y = 5$	$X - Y = 1$	A SOLUTION?
(1; 4)	✓	$-3 \neq 1$	no
(3; 2)	✓	✓	yes
(4; 3)	$7 \neq 5$	✓	no

THE ANSWER IS A PAIR, NOT A NUMBER

Writing “ $x = 3$ ” alone answers half the question. The solution is (3; 2), and both numbers are needed for the mark.

Solving by drawing

Draw both lines on one grid; the point where they cross is the solution.

EQUATION	AS A FUNCTION	TWO POINTS
$x + y = 5$	$y = 5 - x$	(0; 5), (5; 0)
$x - y = 1$	$y = x - 1$	(0; -1), (1; 0)

The lines cross at (3; 2), and substituting confirms it: $3 + 2 = 5$ ✓ and $3 - 2 = 1$ ✓.

A DRAWING GIVES AN ANSWER, NOT A PROOF

If the crossing falls between grid lines — say at (2.4; 1.7) — the graph cannot say so exactly. That is why the next lesson introduces two algebraic methods.

One, none, or infinitely many

THE TWO LINES	PICTURE	SOLUTIONS	EXAMPLE
different gradients	they cross once	exactly one	$y = 2x, y = x + 1$
same gradient, different intercept	parallel	none	$y = 2x, y = 2x + 3$
same gradient and intercept	one line drawn twice	infinitely many	$y = 2x, 2y = 4x$

THE SAME THREE CASES AS FOR ONE EQUATION

A single linear equation had one root, no roots, or every number as a root. A system of two has one pair, no pair, or every point of a line. The pattern repeats one dimension higher.

02 Worked examples

EXAMPLE 1 Is $(3; 2)$ a solution of $\{x + y = 5; x - y = 1\}$?

① $3 + 2 = 5$ ✓

② $3 - 2 = 1$ ✓

③ Both hold, so yes.

ANSWER

Yes

EXAMPLE 2 Solve $\{x + y = 5 ; x - y = 1\}$ graphically.

① $y = 5 - x$ through $(0; 5)$ and $(5; 0)$.

② $y = x - 1$ through $(0; -1)$ and $(1; 0)$.

③ They cross at $(3; 2)$.

④ Check both equations ✓

ANSWER

$(3; 2)$

EXAMPLE 3 How many solutions has $\{y = 2x ; y = 2x + 3\}$?

① Both gradients are 2.

② The intercepts differ.

③ The lines are parallel and never meet.

④ No solutions.

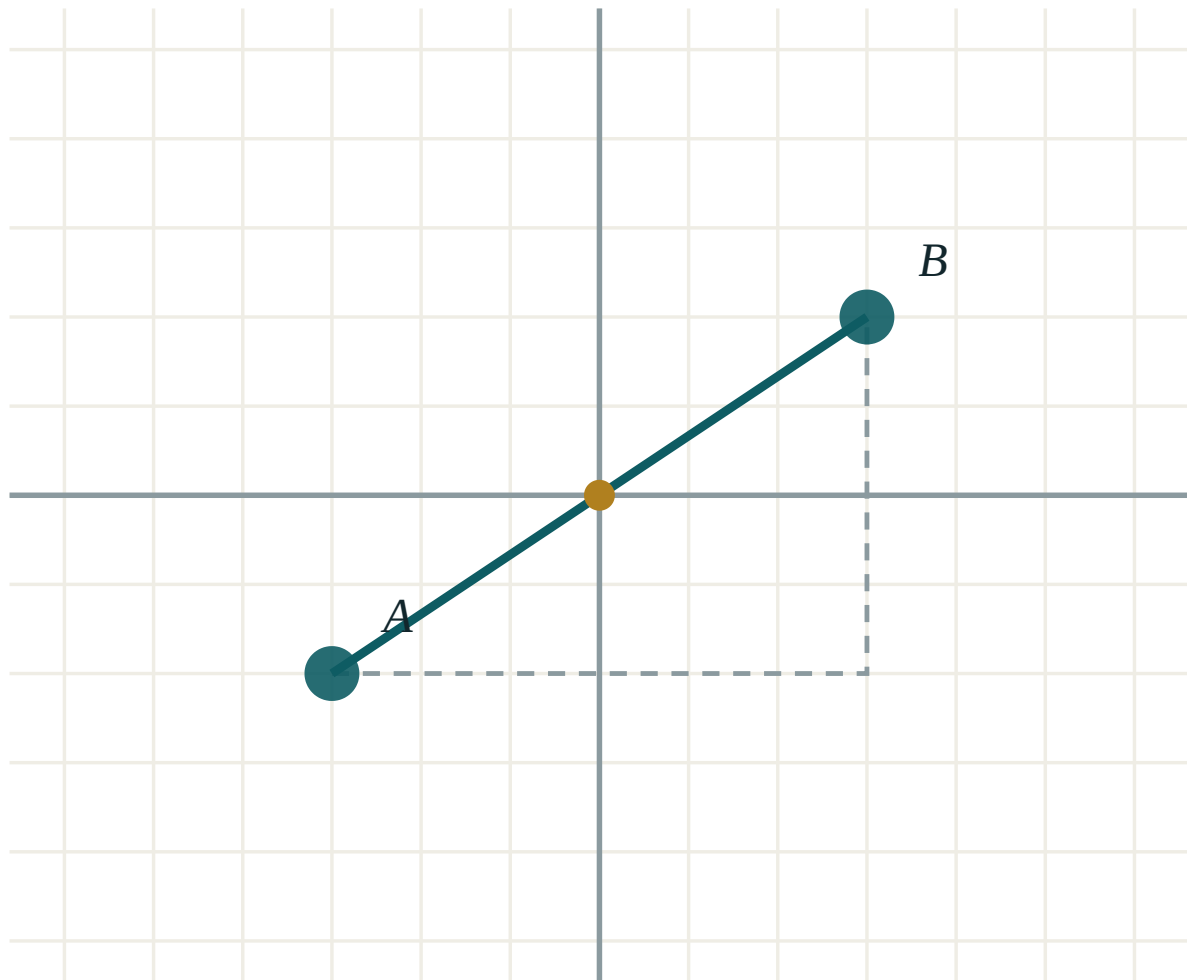
ANSWER

None

03 Interactive model

Ask for five pairs satisfying $x + y = 5$ before anything else; the class produces a line without knowing it, and the second equation then visibly picks one point out of it.

Two lines and their crossing



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
System of equations	Tenglamalar sistemasi	Система уравнений
Two unknowns	Ikki noma'lum	Два неизвестных
Solution of a system	Sistemaning yechimi	Решение системы
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
Graphical method	Grafik usul	Графический способ
Intersection point	Kesishish nuqtasi	Точка пересечения
Parallel	Parallel	Параллельные
Coincident	Ustma-ust tushuvchi	Совпадающие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$x + y = 5$ has:

one solution

no solutions

infinitely many pairs

two pairs

The graph of a linear equation in two unknowns is:

a point

a line

a curve

a region

The solution of a system is:

a number

an ordered pair

a line

an equation

Graphically it is:

the origin

the crossing point

the intercept

the gradient

Parallel lines mean:

one solution

no solutions

infinitely many

two solutions

Coincident lines mean:

one solution

no solutions

infinitely many

none of these

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Three pairs satisfying $x + y = 5$

ANSWER $(0; 5), (2; 3), (5; 0)$

2. Is $(3; 2)$ a solution of $x + y = 5$?

ANSWER Yes

3. Is $(3; 2)$ a solution of $x - y = 1$?

ANSWER Yes

4. Is $(1; 4)$ a solution of the system?

ANSWER No

5. $x + y = 5$ as a function

ANSWER $y = 5 - x$

6. $x - y = 1$ as a function

ANSWER $y = x - 1$

7. The solution of the system

ANSWER $(3; 2)$

Medium · the standard question

7 PROBLEMS

1. Solve $\{ y = 2x ; y = x + 1 \}$ graphically

ANSWER $(1; 2)$

2. How many solutions has $\{ y = 2x ; y = 2x + 3 \}$?

ANSWER None

3. How many has $\{ y = 2x ; 2y = 4x \}$?

ANSWER Infinitely many

4. Solve $\{ x + y = 6 ; x - y = 2 \}$

ANSWER $(4; 2)$

5. Solve $\{ y = x ; y = 4 - x \}$

ANSWER $(2; 2)$

6. Two lines with different gradients meet

ANSWER exactly once

7. Why is a graphical answer sometimes only approximate?

ANSWER The crossing may fall between grid lines

Hard · stretch and reason

7 PROBLEMS

1. Solve $\{ 2x + y = 7 ; x - y = 2 \}$

ANSWER $(3; 1)$

2. For which k has $\{ y = kx + 1 ; y = 2x + 3 \}$ no solution?

ANSWER $k = 2$

3. For which b has $\{ y = 2x + b ; y = 2x + 3 \}$ infinitely many?

ANSWER $b = 3$

4. Does $(2.5; 2.5)$ solve $\{ x + y = 5 ; x = y \}$?

ANSWER Yes

5. A system of two equations in two unknowns can have how many solutions?

ANSWER One, none, or infinitely many

6. Solve $\{ x + 2y = 8 ; x = 2 \}$

ANSWER $(2; 3)$

7. Write a system whose solution is $(1; 1)$

ANSWER e.g. $\{ x + y = 2 ; x - y = 0 \}$

07 Homework

Homework — 5 tasks

Always give the answer as a pair, and check it in both equations.

- Write five pairs satisfying $x + y = 7$ and plot them.
- Solve $\{ x + y = 7 ; x - y = 3 \}$ graphically.
- Decide how many solutions $\{ y = 3x + 1 ; y = 3x - 2 \}$ has, and why.
- Decide how many solutions $\{ y = 3x + 1 ; 2y = 6x + 2 \}$ has, and why.
- Check whether $(2; 5)$ solves $\{ 2x + y = 9 ; x - y = -3 \}$.